Course Information

Course Number: DAEN 459
Course Title: Capstone Senior Design Planning
Section: 500
Time / Location: MW 9:10-10:00 AM / ZACH 211, F 9:10-11:10 AM / ZACH 344
Credit Hours: 3 (2 hours of lecture, 3 hours of lab per week)

Instructor

Instructor: Alexander Abuabara
Email: abuabara@tamu.edu
Zoom: https://tamu.zoom.us/my/abuabara
Office: ETB 4013
Office Hours: W 1:00-3:00 PM (or by appointment)

Course Description

First in a two-course sequence for the capstone senior design experience; formation of a senior design team, visitation with the team sponsor, preparation of the groundwork for the project, preparation of the project charter and collection or acquisition of initial set of data; provision of instructions on different aspects of capstone design, including ethics, design constraints, applicable standards, project management, report writing specifications and requirements, and oral and visual presentations.

Learning Outcomes

By the end of the course, students will be able to:

- Define and scope a feasible capstone project with clear goals, objectives, and deliverables.
- Develop and validate project requirements in collaboration with the project sponsor and relevant stakeholders.
- Design appropriate system architecture and infrastructure plans to support project implementation.
- Identify, collect, acquire, and prepare the initial datasets required for the project.
- Evaluate data governance, privacy, security, ethical, and responsible-use considerations relevant to the project.
- Develop a realistic project schedule that includes milestones, dependencies, resources, risks, and mitigation strategies.
- Design and develop an initial prototype or proof-of-concept that demonstrates the feasibility of the proposed solution.
- Prepare and deliver a professional project proposal and presentation that clearly communicates the problem, proposed solution, technical approach, implementation plan, and expected outcomes.

Pedagogical Model

Semester 1 (DAEN 459) = "Design and De-risk"
Students should:

- Define the problem,
- Acquire data,
- Validate feasibility,
- Design the system,
- Produce operational plans.

Semester 2 (DAEN 460) = "Build and Deploy"
Students should:

- Implement,
- Test,
- Iterate,
- Evaluate,
- Present final system.

Why I Teach this Course

This course offers you a unique learning experience. I am always especially excited to teach this course, here is why:

- **You finally get to combine everything that you have learned throughout your entire undergraduate curriculum**. This not only includes the data engineering skills you have gained, but also the professional, cultural, and communication skills that you have developed.
- **This is an opportunity for you to try to solve a problem that matters to you**. This course gives you a lot of freedom and choice in an extensive project. This course can be the first step towards moving to the career of your choice.
- **Building a system from start to finish is fun and rewarding**. Students often tell me that this was the greatest experience of their entire college career.
- **Working on teams can be messy, but it's worth it**. It is amazing what you can accomplish together, so much more than on your own. Working on real problems with real people has its challenges, but it is also very rewarding when you successfully complete your goals.

- **Every semester is different, every team is different, every solution is different**. There is always something new to learn and innovate.
- **Smaller class sizes and weekly meetings allow you to connect** with your classmates and your instructor(s).

Course Expectations

As a data engineering student, you are expected to apply the skills and knowledge developed throughout your academic career toward the planning and development of a substantial, real-world data-centric project. These expectations mirror professional data engineering roles, preparing you for collaborative, cross-functional work. You are expected to:

- Contribute actively to your team with clear communication, accountability, and collaboration,
- Attend all meetings, meet deadlines, contribute to team decisions, and resolve conflicts professionally,
- Work with stakeholders to define and refine technical and business requirements,
- Develop a detailed project plan with timelines, data needs, and deliverables,
- Address ethical, privacy, and risk considerations,
- Identify, acquire, clean, and validate datasets, including the elaboration of data dictionaries and relevant metadata,
- Design a robust data architecture, including storage, pipelines, and processing frameworks,
- Document data flow, system architecture, and all planning and data engineering decisions with supportive analysis,
- Present project components clearly to both technical and non-technical audiences,
- Deliver a comprehensive design proposal with diagrams, models, and a clear roadmap,
- Track progress through milestone reviews and maintain an updated project board,
- Manage scope, time, and resources, adjusting plans as needed,
- Apply best practices in version control, reproducibility, and testing,
- Ensure compliance with industry standards, regulations, and school policies, and
- End the course with a feasible, well-scoped project ready for implementation.

Recommended/Reference Textbooks

Available free online at TAMU Libraries:

- **Design of Everyday Things**. D. Norman (2013)
- **Universal Principles of Design**. W. Lidwell, K. Holden, J. Butler (2023)
- **Tragic Design: The Impact of Bad Product Design and How to Fix It**. J. Shariat, C. Savard Saucier (2017)
- **Fundamentals of Data Engineering**. J. Reis, M. Housley (2022)
- **Designing Data-Intensive Applications**. M. Kleppmann (2017)
- **The Data Warehouse Toolkit**. R. Kimball, M. Ross (2013)

Grading Policy

- Graded items and weights:

	Deliverable		**Weight**
1	Professionalism and participation (including peer-review)	Individual	11%
2	Team agreement	Individual	2%
3	Sponsor meeting notes	Individual	5%
4	Problem statement and project objectives	Team	7%
5	Draft project charter	Team	2%
6	Ethics, privacy, and risk assessment	Individual	10%
7	Data requirements and acquisition plan	Team	10%
8	Initial data inventory and profiling report	Team	4%
9	Preliminary system/data architecture	Team	10%
10	Requirements and constraints matrix	Team	7%
11	Work breakdown structure and project schedule	Team	10%
12	Risk register and mitigation plan	Individual	5%
13	Evaluation and verification plan	Team	5%
14	Draft planning report and presentation	Team	1%
15	Final planning report, charter, presentation, and DAEN 460 transition package	Team	11%

- In addition to the graded items listed above, other submissions will be evaluated on a **satisfactory/unsatisfactory** basis. Details of these requirements and milestones will be posted on the Canvas course page and may include review sessions as well as meetings with sponsors and advisors to discuss strategies.
- Grades will be calculated by totaling the points you earn and comparing them to the max. possible points.
- Grades assigned are A: 90%–100%, B: 80%–89.9%, C: 70%–79.9%, D: 60%–69.9% and F for less than 60%.

Late Work Policy

Due to the pace and structure of this capstone course, late or make-up work cannot be accommodated. Late or missed assignments will receive a zero. University-excused absences apply only to attendance and in-class activities and do not automatically extend assignment deadlines; any alternative arrangements must be made before the deadline. Grade concerns must be raised within one week of the graded work being returned. Regrade requests will result in the entire submission being regraded.

Course Assignments

Initial guidelines are in the appendix below; more detailed instructions will be shared in class.

Professionalism and participation means being responsible, prepared, and actively involved throughout the project. This includes attending meetings, completing assigned tasks on time, communicating clearly with teammates and instructors, accepting feedback, and contributing ideas to solve problems. Professionalism also means producing accurate and organized work, following ethical practices when handling data, and using appropriate technical tools and documentation. Strong participation shows that a student is dependable, works well with others, and contributes consistently to the success of the project.

All submissions and presentations are formal, graded deliverables and must be uploaded to Canvas as assigned. All team members must participate, with leadership rotating across the team. Teams should hold a brief debrief after presentations. The course concludes with a final poster presentation of each team's capstone project plan. All DAEN capstone teams participate.

Teams are formed by the instructor based on preferences or direct project assignment. Each team has an ISEN faculty advisor, with the instructor guiding the course and the sponsor serving as the client.

> *Faculty Advisor* – Each team must have a faculty advisor from the ISEN department, and the instructor can suggest advisors who fit the project. The advisor is your subject matter expert, offering guidance and suggestions along the way. Teams should meet with their advisor every two weeks unless the advisor approves a different schedule. Faculty advisors are actively involved and considered part of your team, so treat them with respect and courtesy.
>
> *Additional Faculty Advisor* – Teams can add a secondary advisor after checking with the course instructor. These advisors can be from any Texas A&M department or affiliated organization. If you want an advisor from outside Texas A&M, you'll need written approval from the instructor. The request should include their name, email, phone number, organization, area of expertise, and a short explanation of why their help would be valuable.

Course Material and Copyright

All documents used in this course are copyrighted. Here, "documents" means all materials generated for this class, including, but not limited to, course syllabi, course notes, quizzes, exams, problem sets, slide sets, and all materials appearing on the Canvas website or sent to you via email. Because these materials are copyrighted, you do not have the right to copy any of them for any purpose other than your own personal academic use unless the instructor expressly grants permission. Course materials may not be given or sold to any for-profit enterprise.

Confidentiality and Non-attribution

In this course, we will often use client data, project details, or conversations that need to stay within our group. You can share this information with classmates and the teaching team, but not outside the course. To keep things confidential, store notes or files in a safe way (like a folder, binder, or password-protected device) and keep them with you or in a secure place (room, backpack, car). Non-attribution means focusing on ideas, not people. Instead of saying "*José said Range Rovers have reliability issues*," say "*I understand that Range Rovers have reliability issues*." <u>Avoid tying insights to individuals or teams</u>.

Attendance & Class Participation

During the first two weeks, everyone meets during regular class time. Once teams are set, each team will give a technical review about every two weeks, and you're encouraged to attend other teams' sessions to see their work. The instructor may adjust the schedule or add events as needed. Attendance is required for all class sessions, sponsor meetings, and presentation days, unless you have a university-excused absence (see https://student-rules.tamu.edu/rule07).

Team Meetings – Active engagement is a key part of the capstone experience. Students are expected to fully participate in all team activities, including meetings with sponsors, faculty advisors, and internal discussions. These interactions help the project succeed while giving you opportunities to collaborate, solve problems, and practice professional communication skills. Meaningful participation benefits both you and your team throughout the course.

In team projects, everyone is expected to contribute fairly. When someone doesn't, the rest of the team carries the burden, which isn't fair. This problem, often called social loafing, can take many forms:

- Ignoring emails, texts, or calls from the team.
- Being told there's a communication issue but not fixing it.
- Expecting the team to work around your personal schedule.
- Missing meetings, showing up late, or leaving early.
- Submitting work that is incomplete, full of mistakes, or unusable.
- Working on unrelated ideas without team approval, then expecting credit.
- Attending meetings but spending more time on your phone than participating.

What's expected of you:

- Be available, stay in touch with your team. It's your responsibility to make sure communication works.
- Do your part! All must contribute a fair share of the project. Fair doesn't always mean equal. Tasks may vary. Teams are encouraged to assign work that fits members' skills, but everyone must contribute in a meaningful way. Participation (or lack of it) will be reflected in presentations and peer reviews.
- Always show courtesy and respect! Professional behavior is always expected. Teammates are colleagues, use respectful language, and contribute in a way that shows maturity and responsibility.
- If a student consistently fails to participate, faculty may move them into a ***Track B Option***: working alone under direct supervision. In this track, the highest possible grade is a C (if all requirements are met).
- Speak up early! If a teammate still isn't contributing after a TA discussion, let faculty know right away. Problems don't fix themselves, and waiting until the end of the semester leaves no time for solutions.

Project Sponsor Interaction – Teams should meet with the project sponsor's point of contact (POC) about once a month, adjusting as needed for the sponsor's schedule. Meetings can be by phone or video call (email doesn't count). All team members should attend unless there's a university-excused absence or a class conflict. The team and sponsor will coordinate meeting times, and only the sponsor can cancel. If a meeting is canceled or the sponsor feels the schedule isn't working, note it in your technical presentations and discuss a solution with the instructor.

Courses Schedule (Tentative)

DAEN 459

Week	Topics	Team / Lab activities	Milestones / Deliverables
1	Capstone expectations; design process; professional conduct	Review sponsor projects; skills and interest inventory	Individual skills profile and project preferences
2	Team formation; teamwork; roles and responsibilities	Form teams; select projects; establish communication practices	Team agreement and preliminary role assignments
3	Sponsor engagement; stakeholder analysis; effective interviewing	Initial sponsor meeting; identify stakeholders and decision context	Sponsor meeting notes and stakeholder register
4	Problem framing; needs assessment; requirements elicitation	Translate sponsor needs into a preliminary problem statement	Problem statement and project objectives
5	Project charters; scope definition; assumptions and constraints	Define scope, exclusions, success criteria, and major deliverables	Draft project charter
6	Ethics, privacy, security, bias, and responsible data use	Conduct ethical and data-governance review	Ethics, privacy, and risk assessment
7	Data requirements; data provenance; acquisition strategies	Inventory available data; identify gaps and access dependencies	Data requirements and acquisition plan

8	Initial data collection and profiling	Acquire an initial dataset; assess schema, quality, completeness, and suitability	Initial data inventory and profiling report
9	Data engineering architecture and tool selection	Develop source-to-output architecture; evaluate platforms and tools	Preliminary system/data architecture
10	Applicable standards; legal, organizational, and design constraints	Identify standards and convert constraints into requirements	Requirements and constraints matrix
11	Project management; work breakdown structures; estimation	Develop tasks, dependencies, schedule, ownership, and communication plan	Work breakdown structure and project schedule
12	Risk management; feasibility; change control	Assess technical, data, schedule, personnel, and sponsor risks	Risk register and mitigation plan
13	Evaluation planning; alternatives; metrics and acceptance criteria	Define baseline, alternatives, validation methods, and performance measures	Evaluation and verification plan
14	Technical writing; visual communication; design reviews	Integrate planning artifacts; peer and instructor design review	Draft planning report and presentation
15	Professional presentations; transition to implementation	Formal sponsor presentation; revise plans based on feedback	Final planning report, charter, presentation, and DAEN 460 transition package

DAEN 460

Week	Topics	Team / Lab activities	Milestones / Deliverables
1	Project reactivation; execution expectations; configuration management	Reconfirm scope, roles, sponsor expectations, data access, and schedule	Updated project plan and kickoff memorandum
2	Reproducible development; repositories; testing strategy	Configure development environment, repository workflow, and issue tracking	Reproducibility and testing plan
3	Data ingestion and acquisition	Implement source connections and initial ingestion processes	Working ingestion prototype
4	Data quality engineering; validation and observability	Implement schema, completeness, accuracy, and anomaly checks	Data-quality rules and baseline assessment
5	Data transformation; integration; metadata and lineage	Build repeatable cleaning, transformation, and integration workflows	End-to-end pipeline increment 1, PRESENTATION 1
6	Storage, data modeling, performance, and security	Implement storage/modeling layer; review access controls	Architecture and implementation review
7	Exploratory analysis; baselines; analytical validity	Establish descriptive findings and baseline performance	Technical progress report 1
8	Midsemester design review	Demonstrate pipeline and preliminary analysis to faculty and sponsor	Working prototype, Midterm PRESENTATION 2
9	Application of analytical or data engineering methods	Implement primary models, algorithms, or decision-support methods	Analysis/modeling increment 1
10	Alternative designs; trade-off analysis; design constraints	Compare approaches using technical and stakeholder criteria	Alternatives and trade-off analysis
11	Verification, validation, testing, and uncertainty	Perform system tests, sensitivity analysis, error analysis, or model validation	Validation and testing report, PRESENTATION 3
12	Interpretation; recommendations; operational implications	Convert results into sponsor-relevant findings and recommendations	Draft findings and recommendations
13	Deployment, maintainability, documentation, and handoff	Package pipeline, code, dashboard, model, or other product for sponsor use	Release candidate and technical documentation
14	Report writing; presentation design; peer review	Conduct final technical review and rehearsal; resolve remaining defects	Draft final report, PRESENTATION 4

15	Final design presentation and project handoff	Present results; demonstrate product; transfer artifacts to sponsor	Final report, showcase presentation, product, repository, and sponsor handoff

Tips – (i) Be nice. Be honest. Don't cheat. **Adhere to all University Policies**. (ii) You all are important to me: include the course number in email subject so I can prioritize it. (iii) Every step matter and each build on the next: stay engaged, keep up, ask questions. (iv) Learning takes steady effort: start early, not last-minute cramming. (v) Engineers turn data into meaningful information, practice here, get prepared. (vi) Clear, well-organized work is as important as the content itself. (vii) Use others' ideas to build your own but always credit your sources: good scholarship means giving credit where it's due. (viii) No audio or video recording without permission; it's a conduct violation otherwise.

Technology Support – For technical support contact isen-helpdesk@tamu.edu.

AI Statement

Students are permitted to use artificial intelligence tools as part of their work in this course. AI should be treated like any other academic or professional tool: it may assist with brainstorming, organization, research, revision, or problem-solving, but the student remains responsible for the final work submitted.

Students are responsible for verifying the accuracy, quality, and appropriateness of any information produced with AI. Submitting inaccurate, fabricated, or misleading information generated by an AI tool does not excuse the student from responsibility for the work.

Any ideas, information, quotations, images, or other material drawn from outside sources must be properly cited according to the citation style required for the assignment. When AI has contributed substantially to an assignment, students should also acknowledge or cite their use of the AI tool when appropriate.

AI should support your learning—not replace your own thinking, judgment, or understanding. Students may be asked to explain, discuss, or demonstrate the reasoning behind work they submit. Use AI in a way you can feel confident about and openly acknowledge. If you would be uncomfortable explaining how you used AI, reconsider whether that use is appropriate for the assignment.

Initial guidelines

This capstone course serves as an integrative, high-impact educational experience that challenges students to integrate their accumulated knowledge and apply it to a complex, real-world problem. The capstone projects encourage students to adopt a hands-on approach by building solutions from the ground up rather than relying only on existing tools or templates. The goal is to improve essential skills such as project management, communication, and interdisciplinary analysis, thereby preparing students for professional or academic success. Figure 1 provides a broad overview of how data moves through a system, highlighting the sequential steps of data engineering and the critical underlying disciplines that ensure its success.

Unlike *busy work* which often involves repetitive, low-cognitive tasks with limited connection to meaningful learning outcomes, the capstone project demands critical thinking, research, and collaboration. Here "reinventing the wheel" is not busy work but a deliberate method for gaining insights, building confidence, and preparing for real-world innovation. Drawing inspiration from Richard Feynman's quote, "What I cannot create, I do not understand", these are some of the great reasons to reinvent the wheel:

- Learn how wheels are made
- Teach others about wheels
- Learn about the inventors of wheels
- Be able to change wheels (or fix) when they break
- Build a better wheel (for some definition of better)
- Help someone in need of a very special wheel (maybe for a wheelchair?)
- Learn a tiny slice of what it means to build a larger system (such as a vehicle)

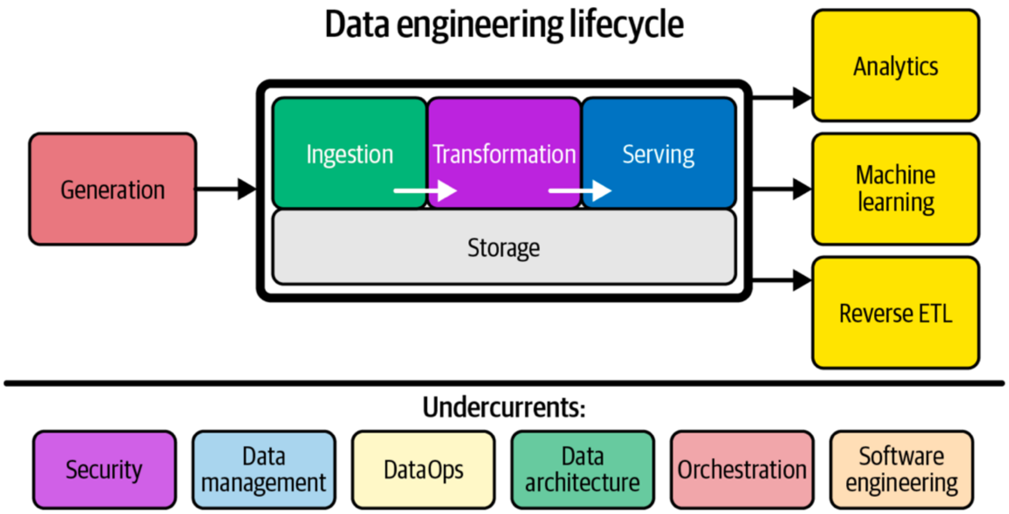

*Figure 1 – Components and undercurrents of the data engineering lifecycle. Image extracted from **Fundamentals of Data Engineering: Plan and Build Robust Data Systems**, J. Reis, M. Housley, 2022.*

V-model

The V-model (Fig. 2) provides a structured framework ideal for planning the Data Engineering Capstone Senior Design courses. It guides students from defining project goals and requirements to detailed system design and implementation. This model promotes a detailed and careful planning, iterative development, and continuous verification, helping students understand the full lifecycle of a data engineering system, from concept and design to deployment and maintenance. Each phase on the left is mirrored by corresponding testing and validation on the right, ensuring alignment between design and outcome.

Within this framework, a Project Charter is typically the first formal document, authorizing the work, defining objectives, scope, stakeholders, and approach, and giving each team the authority to access necessary resources and tools. Once the charter is approved, the Project Management Packet is assembled, detailing schedules, risks, communications, and other operational plans, and ensuring execution stays consistent with the V-model. Like a movie, the charter is the *green light*; the packet is the *production plan*. Below you have more details about the Project Charter and the Project Management Pack.

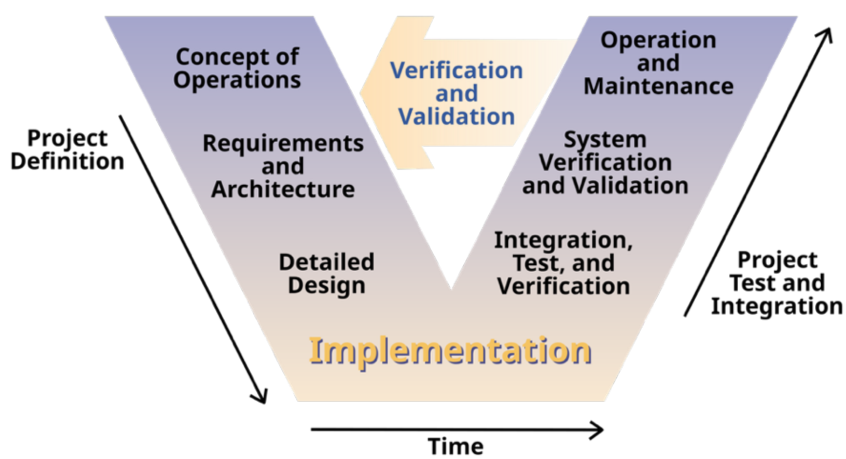

*Figure 2 – Systems Engineering Process. Image extracted from **Clarus: Concept of Operations, Federal Highway Administration (FHWA)**, 2005. (Publication No. FHWA-JPO-05-072, authors: L. Osborne, J. Brummond, R. Hart, M. Zarean, S. Conger.)*

1. Project Charter

The Project Charter is a concise, authoritative document that formally authorizes a project. In a two-semester capstone sequence, the final sponsor-signed charter is the *green light* to proceed to implementation in the second capstone semester. Endorsed by the project sponsor, it grants the project manager and team the authority to commit organizational resources and establishes a common understanding of the project's purpose, scope, and constraints. Treat the charter as both a contract and a living reference; if conditions change, revise it formally rather than drifting informally. These are important aspects that need to be addressed in the charter:

1. **Authority & accountability**: Sponsor sign-off clarifies who owns vision and acceptance.
2. **Unified vision**: Forces the team to articulate a shared understanding before deep technical work.
3. **Baseline for changes**: Scope or resource changes must be justified against the charter, avoiding unpleasant surprises.
4. **Early risks**: Listing assumptions and constraints reveals potential (ethical/legal/technical) risks early enough to plan.
5. **Communication tool**: Because it is short, executives and non-technical stakeholders will actually read it.

2. Project Management Packet

The Project Management (PM) Packet is a single living document that guides a data-engineering capstone team from idea approval through final hand-off. Think of it as the project's constitution that captures what you will build, how, by whom, and when so faculty, sponsors, and teammates always have one authoritative source of truth. Keep the packet lean (about 10 pages) and version-controlled, and revisit it whenever scope, risks, or constraints change. A well-maintained packet not only streamlines your work, but it also showcases professional project-management discipline to future employers. How it's used:

1. **Proposal phase**: Submitted with your charter to secure faculty and sponsor sign-off.
2. **Execution phase**: Updated after each sprint review; serves as the agenda for weekly stand-ups.
3. **Evaluation phase**: Forms the backbone of your final report and memo, proving objectives were met.

Key components of the PM package and their functions:

- **Work Description**: Plain-language summary of what the team will build or analyze.
- **Deliverables**: Defines concrete outputs (code, datasets, report, poster) to hand over.
- **Project Boundaries**: States what is in and out to guard against scope creep.
- **Acceptance Criteria**: Minimum conditions the sponsor must see to sign off.
- **Constraints**: Non-negotiable limits (tools, data privacy, semester length, etc.).
- **Assumptions**: Conditions taken as true (e.g. data access) that may later change.
- **Quality and Performance Metrics**: How accuracy, reliability, and efficiency will be judged.
- **Metrics & Verification Methods**: How each metric will be measured, tested, and logged.
- **Timeline and Milestones**: Semester checkpoints that trigger progress reviews and pivots.
- **Task Priority**: Ranks tasks so critical-path items are tackled first.
- **Risks & Risk Management Plans**: Lists top pitfalls and the team's mitigation / contingency moves.
- **Cost Estimates and Budgeting**: Forecasts hours, cloud credits, and purchases to stay on budget.
- **Resources**: People, software, hardware, and data the project will rely on.